



Investigation of water transport around the root for a plant using X-ray imaging technique

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ABSTRACT

Water transport in the root and soil interface has been investigated to reveal water uptake phenomena through roots. However, the mucilage-filled pore space used for water uptake has not yet been fully understood because it is difficult to analyze the multi-phase flow in an opaque porous media. In this study, the variations of water content (WC) in the region near the root of a *Spathiphyllum wallisii* were systematically investigated using an X-ray imaging system. Based on the spatial variations of WC around the root, the water transport phenomena in the mucilage-filled pore space were observed. To explain the water transport phenomena through the porous media around the root, the spatial variations of WC along the vertical direction were also analyzed. The present X-ray imaging technique would be helpful to understand water transport near roots of a plant.

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1. Introduction

Water uptake through the roots of a plant has been investigated for several decades because it has a great scientific importance and engineering applications in fields of agriculture, water management, and phytoremediation [1–4]. The removal of particulate matter (PM) is also another important engineering application. The active transpiration of air-purifying plants has recently been utilized to remove PM because it is helpful to diminish the concentration of PM particles in atmosphere [5]. Considering the fact that the enhancement of water uptake through roots increases the leaf transpiration of plants [6], a study on the water uptake enhancement is highly recommended to remove PM particles effectively. In addition, the water uptake phenomenon has a great influence on the survival strategy of plants living in severe environmental conditions because they have to either acquire the necessary amount of water from the soil or minimize the loss of water [7]. Therefore, a detailed understanding of the water transport phenomena through roots is an important issue to resolve many scientific and engineering problems.

However, the water uptake process near roots has not been fully elucidated because of complex interactions between the surrounding soil and roots [1,8]. Therefore, many studies have been conducted to reveal the water uptake process near roots from a

porous soil. The narrow region near roots, which is called rhizosphere, has an important role in the water uptake from the soil. Mucilage, a polymeric gel released from the root tip, is distributed in the rhizosphere. It holds soil particles, and the interaction between mucilage and soil particles affects the water management of a plant, especially the water uptake process [9,10]. This implies that mucilage plays an important role in water transport in the rhizosphere. However, depending on plant species, soil characteristics, and experimental conditions, mucilage exhibits different physical properties such as hydrophobicity and viscosity [11,12]. The water uptake characteristics also depend on the drying, wetting, and rewetting processes [13]. Moreover, the multi-phase flow caused by the interaction between soil and mucilage makes it difficult to predict the water transport phenomena near the root system. Therefore, experimental studies are highly necessary to understand the water uptake phenomena with varying experimental parameters such as species, soil condition, and mechanical properties of mucilage.

To investigate the water transport through roots and soil, many experimental techniques such as X-ray tomography [14], MRI [15], neutron tomography [16], fluorescence microscopy [17], and light sheet microscopes have been employed [18]. Although these measurement techniques have their own advantages in the study of water transport near roots, the high spatial resolution is essentially recommended to observe the water transport phenomena in the mucilage-filled pore space at the initial stage. Moreover, because the image acquisition time for acquiring 2D X-ray image is

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relatively short, the water transport phenomena can be observed at a short time interval. Recently, a 2D time-resolved X-ray imaging technique with a high spatial resolution has been employed to investigate water transport through porous media [19] and drug delivery inside subcutaneous tissues [20,21]. Thus, in this study, the X-ray imaging system was utilized to directly measure the temporal and spatial variations of water content (WC) to examine the water transport phenomena near roots.

The purpose of the present study is finding water transport phenomena near the root of *Spathiphyllum wallisii* by using X-ray imaging technique. The multi-phase flow phenomenon is highly expected at the initial stage of redrying phase. The transition region which is observed in the present study exhibits short time at the initial stage of redrying phase of *Spathiphyllum wallisii*. Moreover, the area of transition region is narrow. Therefore, X-ray imaging technique with high spatial resolution was employed to demonstrate water transport phenomena near the root. In this study, the WC information was quantitatively analyzed to investigate the water transport phenomena near roots. The WC gradient values in unit time were also analyzed to explain water transport in each region. Finally, the differences in WC values between each region in the vertical direction were obtained to examine the effects of gravity and viscosity on water transport.

2. Material and method

2.1. X-ray imaging

Fig. 1(a) shows the experimental setup of the 2D X-ray imaging system used in this study. An X-ray CCD camera (Hamamatsu, C9300, Japan) was employed to acquire X-ray images, and an X-ray source (Varian A272) was utilized to generate a focused X-ray cone beam. When the X-ray cone beam was passed through an optical device mounted in front of CCD camera, the X-ray beam changed into visible light. The optical device was composed of a scintillator and a fiber optic plate. To synchronize the X-ray source and CCD camera, a delay generator was utilized. The distance from the test plant to the CCD camera was 60 cm, and that from the CCD camera to the X-ray source was 100 cm. The effective pixel size of the CCD camera was $9\ \mu\text{m} \times 9\ \mu\text{m}$, and the corresponding field of view (FOV) was $36.0\ \text{mm} \times 24.0\ \text{mm}$ in physical dimension. The

optimal X-ray exposure conditions set for this study were 80 kVp and 100 mA.

2.2. Sample preparation

A sample of *Spathiphyllum wallisii* was bought from a local market. Fig. 1(b) shows an optical image of *Spathiphyllum wallisii* sample tested in this study. It was grown in a soil collected under a conditioned environment (relative humidity: $30 \pm 2.5\%$, average temperature: $24 \pm 1.5\ ^\circ\text{C}$) and lighting [13 h/day, $250\ \mu\text{mol m}^{-2}\ \text{s}^{-1}$ photosynthetically active radiation (PAR)].

2.3. Experimental apparatus and method

Sand was sieved using a meshed sieve shaker, and the diameter range of the test sand samples was $150\text{--}200\ \mu\text{m}$. Next, the *Spathiphyllum wallisii* sample was placed in a container of $310\ \text{mm (H)} \times 235\ \text{mm (V)} \times 40\ \text{mm (depth)}$ in physical dimension. After placing the plant in a vertical position, the container was shaken to pack the sample stable. The plant sample was watered up to the saturated condition. Fig. 2 shows the schematics of the experimental procedures of this study. At the end of the drying phase (Day 4), the water was supplied until the water was saturated. We called this phase the forced rewetting phase. A syringe pump was utilized to slowly inject water in the region near the root, and the injection flow rate was $25\ \mu\text{L min}^{-1}$. After the injection process, the plant sample absorbed water from the soil, and then the redrying phase began. Temporal variations in WCs were analyzed during the redrying phase to reveal the water transport phenomena near the root. During this phase, X-ray images were captured consecutively with time intervals of 10 s, and the exposure time for each image was 1 s. The time interval was determined based on the variation of WC.

2.4. Image processing and quantification of water content

To demonstrate the water transport phenomena near the root, the quantitative information of WC was acquired by employing the X-ray imaging technique. The WC information acquired from both sides of the root was averaged statistically. To reduce the anode heel effect of 2D X-ray images, the flat field correction

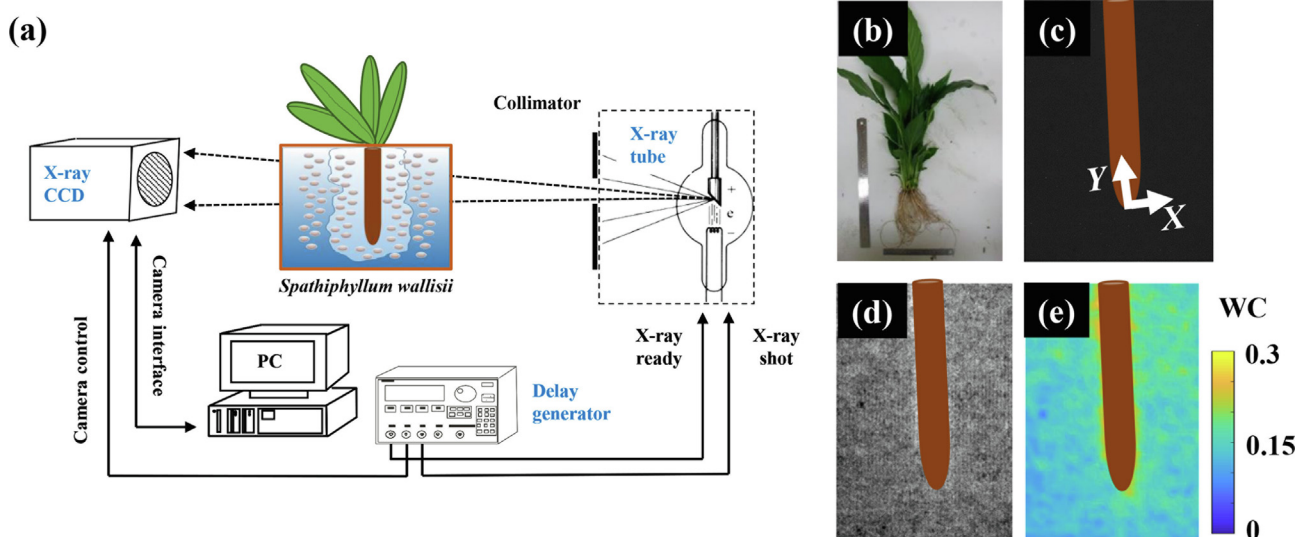


Fig. 1. (a) Schematic diagram of the experimental setup. (b) An optical image of *Spathiphyllum wallisii* sample tested in this study. (c) Typical X-ray raw image, (d) processed image and, (e) WC distribution near the root.

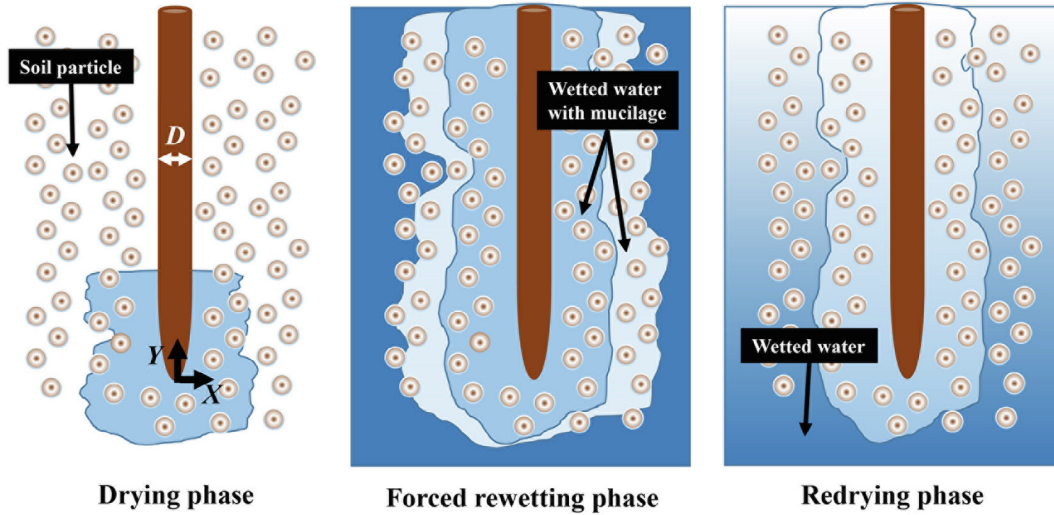


Fig. 2. Schematics of the experimental procedures of this study.

(FFC) method was employed, and the image qualities of X-ray images were modified accordingly [19]. The FFC can be expressed as follows:

$$\text{FFC image} = \frac{\text{Object image} - \text{Offset image}}{\text{Gain image} - \text{Offset image}} \quad (1)$$

Here, the offset image is the image acquired without X-ray beam illumination, and the gain image is the image obtained without the test sample. Fig. 1(c) and (d) shows the typical raw X-ray image and processed image near the root, respectively.

To estimate the water thickness x at an arbitrary position, the following Beer-Lambert law was employed:

$$I = I_0 \cdot \exp(-\mu \cdot x), \quad (2)$$

where I and I_0 indicate the transmitted and incident intensities of X-ray beam, respectively. μ is the linear absorption coefficient of water. The intensity values were acquired from the identical pixel position of the gain and objective images to obtain the WC information at each location. Based on equation (2), the WC information (Fig. 1(e)) along the X-ray beam was acquired from intensity values as follows:

$$\text{WC} = \frac{\ln I - \ln I_s}{\ln I_w - \ln I_s}, \quad (3)$$

where I_s and I_w are the intensities when the container was filled with soil and saturated water in soil, respectively. I indicates the intensity value of the water-soil mixture. For the calibration, the WC information at each water thickness is determined according to the average intensity value for the 5 images. The intensity variation in X-ray images was calibrated with varying water thickness of water jacket with the range of 0–2 mm. A calibration curve ($I = I_0 \cdot \exp(-0.1082 \cdot x)$) was obtained using curve fitting on the variations in gray level with respect to the water thickness and corresponding R^2 value is 0.9967. Therefore, water contents information was obtained by using a calibration curve.

3. Results and discussion

3.1. Variation of water content near the root

When the root begins to absorb water from the soil (redrying phase), the root soaks up water through the tip, where new rhizosphere is formed. Therefore, the tip of the root was determined as a

region of interest (ROI) to observe the water uptake phenomena. The measured WC values exhibited different spatial distributions with the lapse of time. At first, the variations in WC values along the horizontal direction were analyzed during the redrying phase to understand the complex multi-phase flow near the root. Because the initial stage of redrying phase is short, the X-ray imaging technique whose image acquisition time is less than 10 s was employed to investigate water transport near root.

Fig. 3 shows the temporal evolutions of WC values along the X/D direction. The WC value in X/D position was obtained by taking an average of WC values along the vertical direction in the range of $0 < Y/D < 5.63$. The average WC values were quantitatively analyzed as functions of X/D position and time after water supply. The position of ($X = 0, Y = 0$) was denoted as the center position of the tip of the root. D indicates the diameter of the root, and the mean diameter is approximately 2.01 mm. Therefore, the position of $X/D = 0.5$ indicates the interface between the root and soil. When the root diameter is 2 mm, the rhizosphere region is approximately 1.5 mm ($X/D = 1.25$) [22]. As a result, WC values decreased gradually at all X/D positions as the time increased. At the time of water injection (0 min), the WC values were similar in the region where X/D was less than 1.0625. However, beyond the X/D position

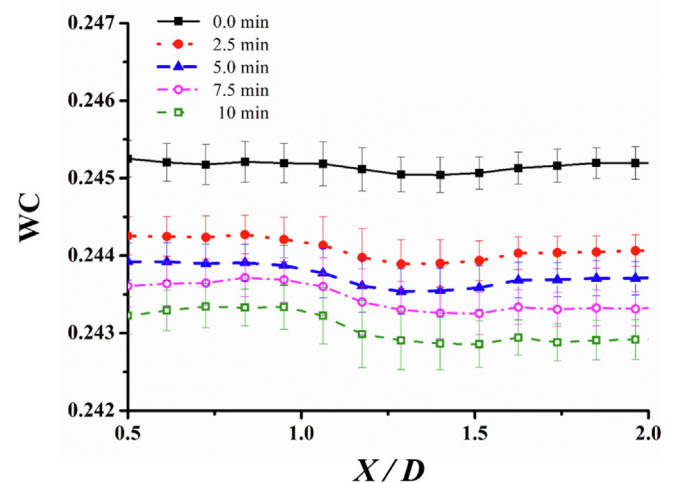


Fig. 3. Temporal evolutions of water content along the horizontal distance X/D from the root.

of 1.0625, the WC values oscillated slightly and approached a constant value up to the X/D position of 1.625. When X/D was larger than 1.625, the WC values maintained a constant value. The WCs values are 0.2453, 0.2450 and 0.2452 at X/D positions of 0.5, 1.4, and 2.075, respectively. As the time increased, this tendency reduced gradually. During the redrying phase, the WC values in the rhizosphere region were larger than those in the bulk region. This implies that the root adjacent to the soil contains more water than the bulk soil region.

To explain the differences in the water transport phenomena along the X/D direction, the spatial variations of WC gradient were evaluated. Fig. 4 shows the spatial variations of WC gradient along the X/D direction as time increases. The variations of WC gradient were classified into three regions, namely, $X/D < 1.0625$ (rhizosphere region), $1.0625 < X/D < 1.625$ (transition region), and $1.625 < X/D$ (bulk region). In this study, the transition region newly defined to explain the complex flow phenomena in the region near the root. The WC gradient exhibits local minimum and maximum peak values at specific locations. Therefore, the beginning and end boundaries of the transition region correspond to the points of local peak values.

In the rhizosphere region, the WC gradient exhibits a decreasing tendency as X/D increases. Compared to the rhizosphere region, the slopes of WC gradient increased greatly in the bulk region. This implies that the spatial variations of WC gradient along the horizontal direction in the rhizosphere are much larger than those in the bulk region. Independent of the elapsed time, WC reaches the maximum value in the rhizosphere region, as illustrated in Fig. 3. At the end of the drying phase, mucilage exhibited hydrophobic characteristics [1]. As mucilage became hydrophobic, the soil started to be slowly rewetted [16]. During the drying phase, the soil and mucilage mixtures were wetted in the rhizosphere area. After supplying water (forced rewetting phase), the interface between the rhizosphere and bulk region would not be filled with water because of the hydrophobic feature of mucilage. They reduced the WC values in the transition region. These differences in WC values of the bulk and transition regions diminished gradually as time increased.

3.2. Water transport in soil around the root

To explain water transport through porous media, Navier-Stokes equation was employed. The equation can be simplified

by assuming the flow to be incompressible and stationary. Darcy's law, which is a simplified equation for creeping flow, can be expressed as follows:

$$q = -K \nabla \Phi, \quad (4)$$

where K is the hydraulic conductivity, q is the volume flow rate, and Φ denotes the pressure. K is a function of fluid viscosity and WC at a given area [8]. The hydraulic conductivity of a bulk soil can be expressed as follows [8]:

$$K_{rhizosphere}(WC, \mu) = \frac{\mu_{water}}{\mu} K_{bulk}(WC). \quad (5)$$

In the redrying phase, as the viscosity of mucilage (μ) increases [11], the hydraulic conductivity in the rhizosphere decreases according to equation (5).

To examine the changes in the hydraulic conductivity at each region, the temporal evolutions of $\Delta(WC)/\Delta t$ were evaluated [Fig. 5(a)]. $\Delta(WC)/\Delta t$ indicates the variations of WC in a unit time of 10 s. The absolute values of $\Delta(WC)/\Delta t$ decrease gradually with the lapse of time. The simultaneous decrease in both WC and $\Delta(WC)/\Delta t$ values implies the decrease in water transport during the redrying phase. The mean absolute values of $\Delta(WC)/\Delta t$ in the rhizosphere region are relatively small compared to those in the bulk region tested in this study. This indicates that variations of water quantities in the rhizosphere region is less than that in the bulk region. This phenomenon is attributed to the change in the viscosity of mucilage. Therefore, water transport was diminished as hydraulic conductivity decreases in the rhizosphere region.

Fig. 5(b) shows the temporal variations of the water content ratio WC_{Tr}/WC_{Rh} between the transition region and rhizosphere region. Here, WC_{Tr} indicates the average value of WC in the transition region, and WC_{Rh} indicates the average value of WC in the rhizosphere region. The WC ratio decreases gradually with the lapse of time during the redrying phase. During the initial stage of the redrying phase, the value of WC_{Tr}/WC_{Rh} decreases dramatically. After 2.5 min, the decreasing slope is mitigated for up to 10 min.

Fig. 6 shows the relation between $\Delta(WC)/\Delta t$ and degree of saturation S for the rhizosphere and transition regions. The relation can be approximated by a linear fit. Here, the degree of saturation S is defined as follows [23]:

$$S = \frac{WC - WC_r}{WC_s - WC_r}, \quad (6)$$

where WC_s and WC_r are the saturated and residual WC values, respectively. WC_s was determined as the WC value in the root-soil interface at the initial instant of water injection (0 min) and WC_r was assumed as 0. The degree S affects the permeability and hydraulic conductivity of the porous media [24]. When S increases, the absolute values of $\Delta(WC)/\Delta t$ increase in both rhizosphere and transition regions. For the same value of S , the absolute values of $\Delta(WC)/\Delta t$ in the transition region are larger than those in the rhizosphere region. As water transport, which is a function of $\Delta(WC)/\Delta t$ and WC, is strengthened by reducing viscosity, the viscosity of mucilage would be decreases when moving along the X/D direction. In addition, the amount of mucilage also decreases gradually with increasing distance from the root-soil interface [13]. For this reason, the viscosity, hydraulic conductivity, and water transport vary with the amount of mucilage under the same degree of saturation S . Assuming that the physical and chemical properties of mucilage itself are not changed during the redrying phase, the viscosity of mucilage seems to be a function of the ratio of the amount of wetted water to the amount of mucilage. Therefore, the ratio of viscosities between the rhizosphere and transition regions would be estimated by the ratio of the water transport information of two regions. The corresponding viscosity ratio between the rhizosphere and transition regions is approximately 1.4224 ± 0.265 . In the same

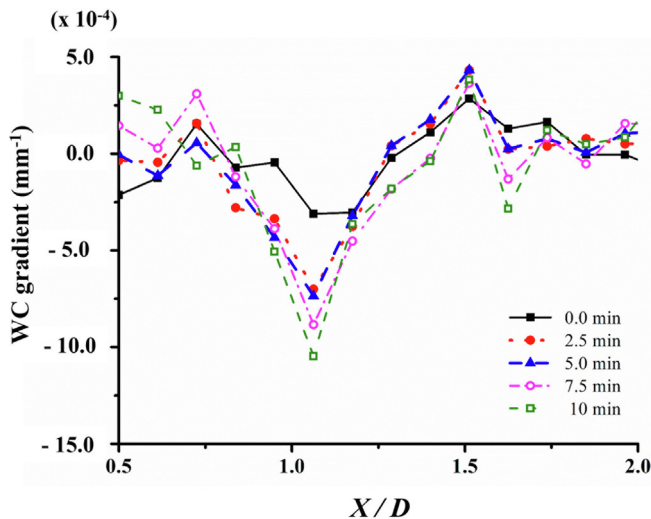


Fig. 4. Temporal variations of water content gradient along the horizontal distance X/D from the root.

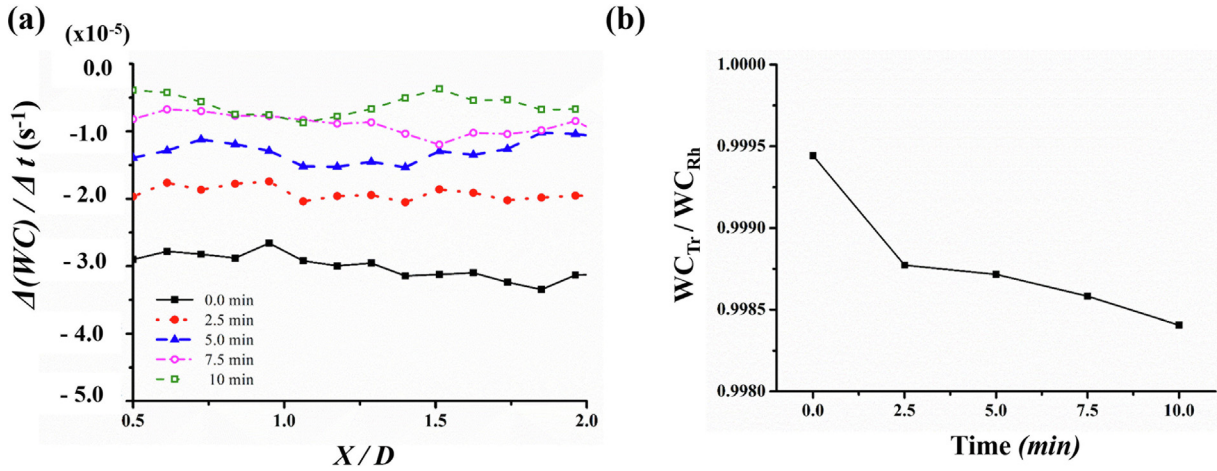


Fig. 5. (a) Temporal evolutions of $\Delta(WC)/\Delta t$ according to distance from root. (b) Temporal variations of water content ratio WC_{Tr}/WC_{Rh} between transition and rhizosphere regions.

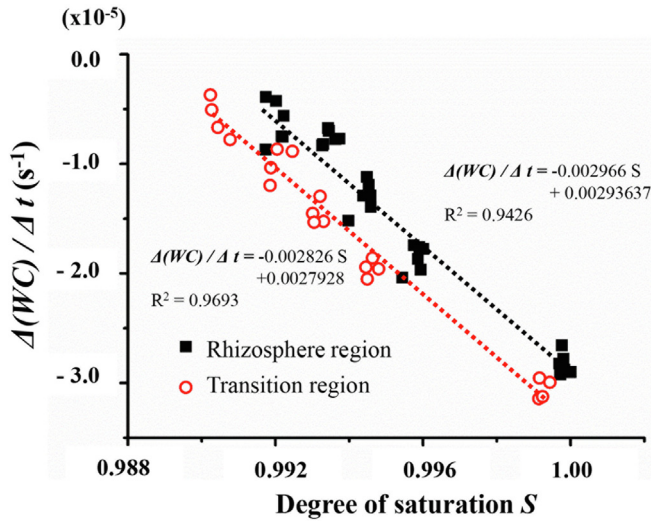


Fig. 6. Relation between $\Delta(WC)/\Delta t$ and degree of saturation in the rhizosphere (black square) and transition regions (red circle). The linear lines represent the first-order approximation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

manner, because the variations of $\Delta(WC)/\Delta t$ in the rhizosphere region under the same S condition are relatively larger than those in the transition region, the viscosity variations in the rhizosphere region would be relatively larger compared to that in the transition region. This implies that viscosity in the rhizosphere region would be decreased more rapidly with increasing distance from the root–soil interface, compared to that in the transition region.

3.3. Variation of water content along the vertical direction

The viscosity of mucilage is one of the main factors affecting the water transport near the root. Other factors include capillary and gravity forces. For most problems associated with capillarity, gravity force has been considered as the only external force [19]. Therefore, the flow phenomena in the bulk region can be explained by using the capillary force acting on the boundary surfaces. Fig. 7 (a) shows differences in WC values of the rhizosphere and bulk regions (ΔWC_{R-B}) along the Y direction. The WC values in each region were averaged along the horizontal direction. The averaged values were normalized by the ΔWC_{R-B} value at the time of 10 min in the upper region of $Y/D = 5.63$.

At the initial stage of the redrying phase, WC in the tip of the root ($Y/D = 0$) has a positive value. Therefore, the WC value in the rhizosphere region is relatively higher than that in the bulk region. As Y/D increases, ΔWC_{R-B} increases gradually. However, in the redrying phase, ΔWC_{R-B} has a negative value at the tip of the root. In the rhizosphere region, because the mucilage holds wetted water, the variation of WC is relatively smaller than that in the bulk region during the redrying phase. However, due to the gravity force acting on the bulk region, water in the upper region moves toward the lower region. Thus, at the tip of the root ($Y/D = 0$), the WC value in the bulk region is increases gradually as time increases. Consequently, WC in the bulk region decreases when Y/D increases. Therefore, ΔWC_{R-B} increases with increasing Y/D . This phenomenon is further accelerated with the lapse of time. As Y/D increases from 0 to 5.63, the variations of ΔWC_{R-B} are 0.314, 0.759, and 1.190 at the elapsed times of 0 min, 5 min, and 10 min, respectively.

Fig. 7(b) represents the differences in WC values at the rhizosphere and transition regions (ΔWC_{R-T}) along the Y/D position. The values were also normalized by ΔWC_{R-T} at the time of 10 min in the upper region of $Y/D = 5.63$. Water transport in the transition region is influenced by gravity force and viscose effect caused by mucilage. Thus, ΔWC_{R-T} has positive values at all Y/D positions. This implies that the WC values in the rhizosphere region are relatively larger compared to those in the transition region. At the initial stage of the redrying phase, the ΔWC_{R-T} values increase gradually as Y/D increases. This variations tendency is maintained for 10 min. As Y/D increases from 0 to 5.63, the variations of ΔWC_{R-T} are 0.227, 0.122, and 0.315 at the elapsed times of 0 min, 5 min, and 10 min, respectively. These quantities are relatively smaller than those in the rhizosphere and bulk regions. As the amount of mucilage decreases gradually with increasing distance from the root–soil interface [13], mucilage in the transition region becomes smaller than that in the rhizosphere region. Therefore, the wetted water reduces in the transition region because of the combined effects of gravity and viscosity. Thus, in this region, ΔWC_{R-T} has positive values for all cases, and the variations in ΔWC_{R-T} are relatively smaller compared to the results shown in Fig. 7(a). Conclusively, the gravity and viscose effects have dominant influence on water transport in the bulk region and rhizosphere region, respectively. However, their effects mutually affect water transport in the transition region. The X-ray imaging technique has been used to investigate water transport phenomena inside opaque porous media, because it has advantages in terms of high spatial and temporal resolutions. Moreover, the accessibility to a widely

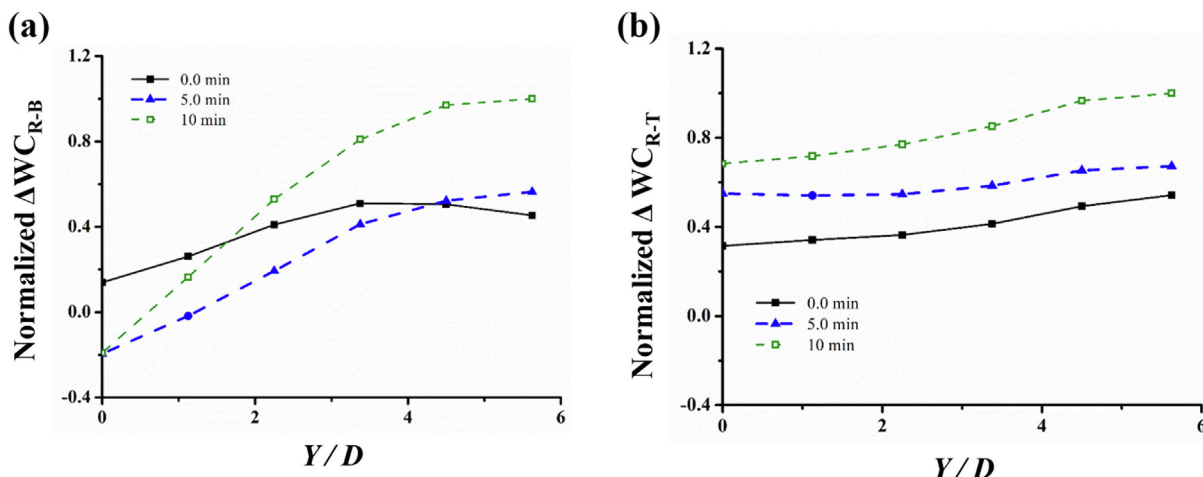


Fig. 7. Temporal evolutions of normalized differences in water content between (a) the rhizosphere–bulk region and (b) the rhizosphere–transition region along the vertical direction.

spread medical X-ray source is relatively easy compared with other non-invasive measurement techniques. However, it is difficult to acquire 3-dimensional information with a wide range of field of view. Thus, a 3D tomographic X-ray imaging technique was developed to get 3D flow information. X-ray dose is another problem in the studies on water transport phenomena inside biological samples. This problem will be solved with the advances of X-ray source and detector. Although the X-ray imaging technique has some technical limitations, it still has great potential to investigate water transport phenomena in porous media.

4. Conclusion

In this study, the spatial distributions and temporal variations of WC in the rhizosphere, transition, and bulk regions were quantitatively analyzed using the 2D X-ray imaging technique. Water transport in the rhizosphere and transition regions is observed to be noticeably influenced by mucilage. Therefore, as gravity, viscosity, and hydraulic conductivity in the wetted water–soil–mucilage mixture influence water transport in the soil, the theoretical analysis of their effects is difficult because of the multiphase flow characteristics in the mucilage-filled pore space. Water transport in the transition region varies depending on the mucilage and pore space distribution. In this study, a transition region was defined to explain the water transport phenomena in the mucilage-filled pore space. Because the transition region exhibits short time at the initial stage of redrying phase, X-ray imaging technique with high spatial resolution was employed. Moreover, the X-ray images were consecutively acquired with short time intervals. The temporal evolutions of WC in unit time were also analyzed to explain the water transport phenomena in each region. Based on the information of WC variations in unit time and the degree of saturation data, the viscosity of mucilage seems to be a function of the ratio of the amount of wetted water to mucilage. Finally, the differences in WC values of three regions along the Y/D direction also demonstrate the effects of viscose force and gravity. The present results will be helpful to understand water transport near roots of a plant.

Author contribution

SYJ, SJL, and HP proposed the study. SYJ and HP performed the experiment and processed the captured images, and then analyzed the experimental data. All authors discussed the results and participated in completing the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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